Polynomial congruences

Finding solutions, if they exist, to problems like

$$f(x) \equiv 0 \pmod{m}$$

where f is a polynomial with integer coefficients (this implied from here on).

Let's first look at existence of solutions for polynomial equations

$$f(x) = 0.$$

Theorem Fundamental Theorem of Algebra. For polys with complex coefficients.

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

where $a_i \in \mathbb{C}$, $a_n \neq 0$ so f has degree n. Then $f(x)$ will have n solutions in $\mathbb{C}$, when counted with ~~multi~~ multiplicities

Q: Does the same hold for poly. congruences?

A: Not so simple or direct.

<u>Ex. 6.1</u> Consider solving

$$2x - 3 \equiv 0 \pmod 4$$

for $x \in \mathbb{Z}_4$. Checking $x = 0, 1, 2, 3$ we see $2x - 3 \equiv 1, 3, 1, 3$

So there's <u>no</u> solutions.

<u>Ex 6.2</u> Consider $x^2 - 1 \equiv 0 \pmod 8$

For $x \equiv 1, 3, 5, 7$ $\quad x^2 - 1 \equiv 1 - 1 \equiv 0 \pmod 8$

For $x \equiv 0, 2, 4, 6$ $\quad x^2 - 1 \not\equiv 0 \pmod 8$

This poly. of degree <u>2</u> has <u>four</u> solutions.

<u>Reminder</u> When we speak of a solution to $f(x) \equiv 0 \pmod{\underline{m}}$ we really mean a congruence

days x modulo m.

eg. $x = 1, 9, 17, 25, \ldots$ all solve

$$x^2 - 1 \equiv 0 \pmod 8$$

These are really all the <u>same</u> <u>solution</u> modulo 8.

Remember previous exercise.

If $a \equiv b \pmod m$ then

$$f(a) \equiv f(b) \pmod m$$

for a poly f with integer coefficients.

<u>6.1</u> Linear congruences, i.e. degree 1 polynomials $f(x) \equiv ax + b \pmod n$.

$$ax + b \equiv 0 \pmod n$$

$$\Longleftrightarrow \quad x \equiv a^{-1} b \pmod n$$

provided a^{-1} exists modulo n

i.e. provided $\gcd(a, n) = 1$

This is theorem 6.1

If $\gcd(a,n)=1$ then $ax \equiv b \pmod{n}$ has a unique solution $x \equiv a^{-1}b \pmod{n}$

Eg. Consider $5x \equiv 3 \pmod{11}$

Check $\gcd(5,11)=1$

So there's a unique solution.

$$x \equiv 5^{-1} \cdot 3 \pmod{11}$$
$$\equiv \underline{9} \cdot 3$$
$$\equiv 27$$
$$\equiv 5 \pmod{11}.$$

by mental checking, or find inverse using Extended Euclidean Algorithm.

$5x + 11y = 1$

$\iff 5x = 1 - 11y$

$\equiv 1 \pmod{11}$

But what does happen if $\gcd(a,n) > 1$.

Can solutions exist? Not always, see ex. 6.1. But they can exist sometimes.

Let's suppose a solution $x \in \mathbb{Z}$ exists for $ax \equiv b \pmod{n}$.

$\exists\, x \in \mathbb{Z}.\quad ax \equiv b \pmod{n}$

$\iff \exists\, x \in \mathbb{Z}.\quad n \mid ax - b$, by def of $\equiv$

$\iff \exists\, x \in \mathbb{Z}\ \exists\, q \in \mathbb{Z}.\quad ax - b = qn.$

$\iff \exists\, x, q \in \mathbb{Z}.\quad b = ax - qn.$

$\iff d \mid b$, where $d = \gcd(a, n)$.

<u>This proves theorem 6.2</u>

If $\gcd(a,n) = d$ then solution(s) exist for $ax \equiv b \pmod{n}$ iff $d \mid b$.

Theorem 6.3 Completes the story for lin. congs.

<u>Theorem 6.3</u>. If $d \mid b$, where $d = \gcd(a,n)$.

then $ax \equiv b \pmod{n}$ has d solutions.

$$t + i\frac{n}{d} \pmod{n}.$$

where t is the unique solution modulo $\frac{n}{d}$

to $\frac{a}{d} \equiv \frac{b}{d} \left(\bmod \frac{n}{d}\right)$.

<u>Sketch of the proof.</u>

$ax \equiv b \pmod{n}$ & $\frac{a}{d} \equiv \frac{b}{d} \left(\bmod \frac{n}{d}\right)$

have exactly the same <u>integer</u> solutions x. since.

$$ax - b = qn \iff \frac{a}{d}x - \frac{b}{d} = q\frac{n}{d}$$

and so the d integers.

$$x \equiv t + i\frac{n}{d} \quad i = 0, 1, \ldots, d-1$$

<u>are</u> all solution to $ax \equiv b \pmod{n}$, so just need to show these are all distinct modulo n.

Continue at 12:00.

Example 6.3

1. Solve $5x \equiv 3 \pmod{24}$.

Apply theorem 6.1 as $\gcd(5, 24) = 1$

so $x \equiv 5^{-1} \cdot 3 \pmod{24}$

$\equiv 5 \cdot 3$ as $5^{-1} \equiv 5 \pmod{24}$

$\equiv 15 \pmod{24}$.

this is the unique solution mod 24.

2. Solve $25x \equiv \underline{\underline{15}} \pmod{120}$

Examine $\gcd(25, 120) =$

$\gcd(5^2, 2^3 \cdot 3 \cdot 5) = 5$

and $5 \mid 15$ so by theorem 6.3

there are 5 solutions modulo 120.

Generated as

$$x \equiv t + i\,24 \pmod{120}$$

where t is the unique solution modulo 24 to the reduced congruence.

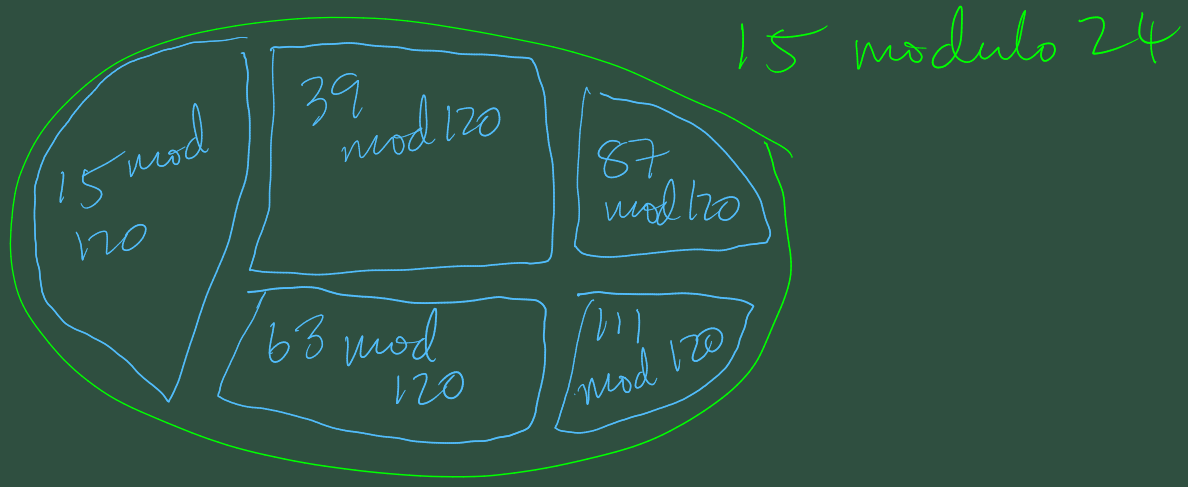
$$5x \equiv 3 \pmod{24}$$

So $t \equiv 15 \pmod{24}$.

and the solutions x are

$$x \equiv 15, 39, 63, 87, 111. \pmod{120}$$

Note on the relationship between congruence classes and moduli



An important tool for solving general poly. congs. $f(x) \equiv 0 \pmod{n}$ is the C.R.T. Chinese Remainder Theorem.

CRT is a tool for "looking a set of congruence classes into a single one"

Theorem 6.4 CRT

Consider the k-congruence classes.

$$(*) \quad x \equiv b_i \pmod{m_i} \quad i = 1, \dots, k.$$

where the moduli $m_1, m_2, \dots, m_k$ are <u>pairwise-coprime</u>, i.e. for any i, j with $i \neq j$ $\gcd(m_i, m_j) = 1$ then the system of simultaneous congruences $(*)$ is equivalent to a <u>single</u> congruence class modulo $M = m_1 m_2 \dots m_k$.

<u>Note</u>: on the "pairwise" term.

the integers $(2, 4, 3, 9)$ are a co-prime 4-tuple, but not pairwise coprime as $\gcd(2,4) = 2$, $\gcd(3,9) = 3$

<u>Proof of CRT</u>. Extremely <u>constructive</u>.

Set up the ingredients/parameters.

$$M := \prod_{i=1}^{k} m_i = m_1 \cdot m_2 \cdot \ldots \cdot m_k.$$

$$M_i := M/m_i = \prod_{\substack{j=1 \\ j \neq i}}^{k} m_j$$

$$M_i' := M_i^{-1} \pmod{m_i}$$

Note this inverse exists since $\gcd(M_i, m_i) = 1$, by the pairwise prime condition on the m_j. $j = 1, \ldots, k$.

The simultaneous solution x to the congruences $x \equiv b_i \pmod{m_i}$ $i = 1, \ldots, k$ is the congruence class.

$$x = \sum_{j=1}^{k} b_j M_j M_j' \quad \underline{\underline{(\bmod M)}}$$

$$= b_1 M_1 M_1' + b_2 M_2 M_2' + \ldots + b_k M_k M_k'$$

<u>Consider this x modulo m_i</u>

$$x \equiv b_i M_i M_i' \pmod{m_i}$$

since each $M_j \equiv 0 \mod m_i$

for each $j = 1, \dots, k$

where $j \neq i$

and $\equiv b_i \pmod{m_i}$ since

$M_i' \equiv M_i^{-1} \pmod{m_i}$

So this $x = \sum_{i=1}^{k} b_i M_i M_i'$

is a solution of each $x \equiv b_i \pmod{m_i}$

Then a last paragraph proving this

is the only solution modulo M. $\blacksquare$

<u>Example 6.4</u>

Q.1. Solve.

$$x \equiv 2 \pmod 5$$
$$x \equiv 3 \pmod 5$$
$$x \equiv 2 \pmod 7.$$

simultaneously.

Note the moduli 3, 5, 7 are pairwise coprime. CRT says there is a unique solution modulo $M = 3 \cdot 5 \cdot 7 = 105$.

<u>Parameters</u>

$M_1 = 5 \cdot 7 = 35$ $\quad M_2 = 3 \cdot 7 = 21$

$M_3 = 3 \cdot 5 = 15$ Their inverses are.

$$M_1' \equiv M_1^{-1} \pmod{m_1}$$
$$\equiv 35^{-1} \pmod 3$$
$$\equiv 2^{-1} \pmod 3$$
$$\equiv 2$$

$$M_2' \equiv M_2^{-1} \pmod{m_2}$$
$$\equiv 21^{-1} \pmod 5$$
$$\equiv 1^{-1} \pmod 5$$
$$\equiv 1 \pmod 5$$

$$M_3' \equiv M_3^{-1} \bmod m_3$$
$$\equiv 15^{-1} \pmod 7$$
$$\equiv 1 \pmod 7$$

So the simultaneous solution x is

$$x \equiv b_1 M_1 M_1' + b_2 M_2 M_2' + b_3 M_3 M_3' \pmod M$$

$$\equiv 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 \pmod{105}$$

$$\equiv 140 + 63 + 30$$

$$\equiv 233 \pmod{105}$$

$$\equiv 23 \pmod{105}.$$

Check: $23 \equiv 2 \mod 3$ ✓

$23 \equiv 3 \mod 5$ ✓

$23 \equiv 2 \mod 7.$ ✓